Karthikeya Kondaveeti

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SUMMARY

Computer Science undergraduate at SRM University AP with a focus on applied machine learning and computer vision, backed by full-stack development experience (MERN). Built projects spanning video-based incident detection, multimodal retrieval-augmented generation, and geospatial forecasting. Currently working on portion-aware food calorie estimation from single images. Seeking a Master's programme in Computer Science, Machine Learning, or Data Science.

EDUCATION

SRM University AP Aug. 2023 – May 2027
B.Tech. Computer Science and Engineering, Amaravathi, India

PROJECTS

Portion-Aware Calorie Estimation from Food Photos *In progress*
Python, PyTorch, Segment Anything, Depth Anything, FastAPI
- Developing a pipeline that segments food items, estimates portion size using monocular depth and a scale reference, and converts it to calories using Indian Food Composition Tables.
- Evaluating against a direct vision-language-model baseline, with a focus on Indian meals and reporting estimates with uncertainty ranges.

Medical Report Explainer — Gemini Pro + RAG
Python, FastAPI, Gemini Pro, ChromaDB, Streamlit
- Built a multimodal assistant that extracts lab values and medications from report images/PDFs into structured JSON using Gemini Pro.
- Designed a retrieval-augmented pipeline over WHO/ICMR guidance and drug leaflets (ChromaDB) so every explanation carries source citations.

Road Incident Detection & Police Alert System
Python, YOLO, MediaPipe, PyTorch, OpenCV, FastAPI
- Built a real-time video pipeline to detect fights and snatching using person tracking, pose features, and a video action-recognition model.
- Designed temporal smoothing and human-verification steps with clip and snapshot alerts to a police dashboard via Telegram/Twilio.

Road Damage & Waterlogging Monitor
Python, YOLO, GeoPandas, XGBoost, Prophet, Folium
- Built a YOLO-based detector for potholes, cracks, and waterlogged patches, with geotagging and road-segment mapping using OpenStreetMap data.
- Designed a rainfall-driven forecasting module (XGBoost/Prophet) to predict deterioration and flooding risk per road segment.

Folio — Online Book Reading Platform GitHub
React, Node.js, Express, MongoDB, Tailwind CSS, Framer Motion
- Built and deployed a full-stack reading platform with JWT authentication, a customizable reader (themes, typography), cross-device progress sync, bookmarks, and highlights.
- Seeded 10 full-text public-domain books and a 35-title catalogue from Project Gutenberg; added Zod validation, rate limiting, an admin console, and unit/integration tests.

EXPERIENCE

Full-Stack Development Intern 2025
Edunet Foundation, Remote
- Developed a full-stack goal tracking application using Node.js, Express.js, and MongoDB.
- Designed RESTful APIs and integrated MongoDB database.
- Built responsive frontend using HTML, CSS, and JavaScript.

SKILLS

Languages: C++, Python, JavaScript, HTML/CSS, Java
ML/CV: PyTorch, YOLO, OpenCV, MediaPipe, XGBoost, Prophet, RAG (ChromaDB)
Frameworks/Technologies: React.js, Node.js, Express.js, FastAPI, Streamlit
Databases: MongoDB, MySQL | **Tools:** Git, GitHub, VS Code, Jupyter Notebook
Spoken Languages: English, Telugu, Hindi, German A1, French A1

CERTIFICATIONS

- Oracle Certified Professional: Java SE 17 Developer — Score: 82%
- MongoDB Certification — MongoDB Associate Developer, MongoDB, 2026
- Full-Stack Development Internship Certification — Edunet Foundation, 2025